CONTROL NARRATIVE DOCUMENT

Pump Station PS-01

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Reviewed by: Process Engineering

1. GENERAL DESCRIPTION

Pump Station PS-01 is designed to transfer process fluid from the suction vessel T-100 to the downstream process unit at a controlled flow rate. The station comprises two centrifugal pumps operating in duty/standby configuration, with automatic changeover on fault detection. The flow is regulated by control valve FV-301 based on the flow controller FIC-201 setpoint.

The system is designed in accordance with ISA-88 batch control standards, ISA-18.2 alarm management requirements, and ISA-84 safety instrumented system guidelines. The Programmable Logic Controller (PLC) shall implement all control logic defined in this document.

1.1 P&ID Schematic

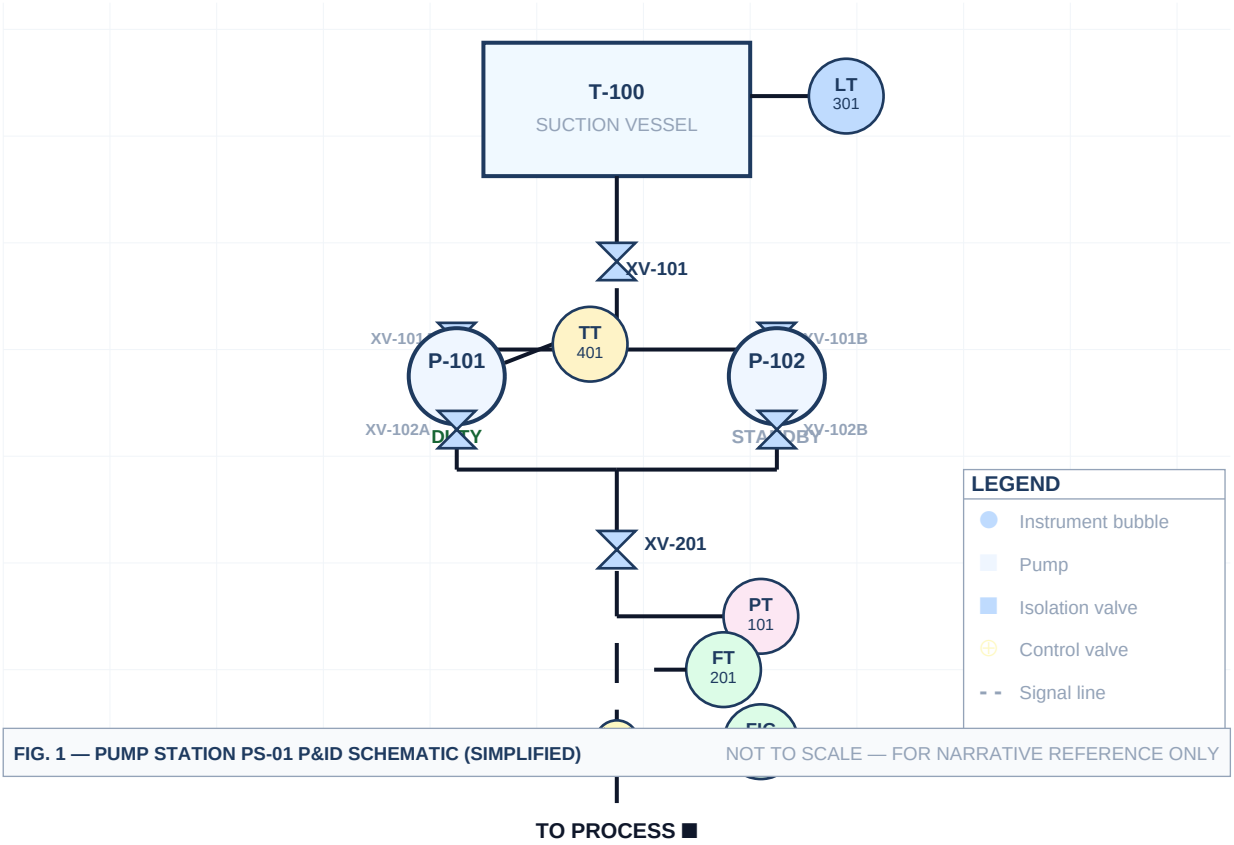

FIG. 1 — PUMP STATION PS-01 P&ID SCHEMATIC (SIMPLIFIED) NOT TO SCALE — FOR NARRATIVE REFERENCE ONLY

2. EQUIPMENT LIST

Tag	Description	Type	Range / Rating	Normal State
P-101	Duty pump	Centrifugal pump	0–150 m³/h, 11 kW	Running
P-102	Standby pump	Centrifugal pump	0–150 m³/h, 11 kW	Stopped
XV-101	Suction isolation valve	Pneumatic gate valve	DN150, PN16	Open
XV-102	Discharge isolation valve	Pneumatic gate valve	DN150, PN16	Open
XV-201	Main discharge valve	Pneumatic gate valve	DN200, PN16	Open
FV-301	Flow control valve	Pneumatic globe valve	Cv 120, fail-open	Auto
PT-101	Discharge pressure	Pressure transmitter	0–300 PSI, 4–20 mA	Active

Tag	Description	Type	Range / Rating	Normal State
FT-201	Process flow	Magnetic flow transmitter	0–200 L/min, 4–20 mA	Active
LT-301	Suction vessel level	Radar level transmitter	0–5 m, 4–20 mA	Active
TT-401	Pump bearing temperature	RTD temperature transmitter	0–150 °C, 4–20 mA	Active
FIC-201	Flow controller	PID controller (in PLC)	SP: 80 L/min	Auto

3. OPERATING MODES

3.1 STOPPED

Both pumps de-energised. XV-201 closed. FV-301 in fail-open position. All alarms monitored but no control action taken. Entry: operator command or ESD activation. Exit: operator initiates STARTUP sequence.

3.2 STARTING

Startup sequence executing. XV-101 opening sequence in progress. P-101 starting. XV-201 opening. Duration: up to 120 seconds. Exit: all permissives confirmed → RUNNING.

3.3 RUNNING

P-101 running (duty). FIC-201 controlling FV-301 to maintain 80 L/min setpoint. P-102 on standby auto-start. All interlocks active. Exit: operator initiates SHUTDOWN or fault triggers ESD.

3.4 STANDBY

P-101 stopped due to fault. P-102 auto-started. FIC-201 continues control. Alarm generated for P-101 fault. Exit: operator acknowledges fault, manually restores P-101.

3.5 STOPPING

Shutdown sequence executing. FV-301 closing. P-101 stopping. XV-201 closing. Duration: up to 60 seconds. Exit: sequence complete → STOPPED.

3.6 EMERGENCY SHUTDOWN

All pumps stopped immediately. XV-201 closed. FV-301 de-energised to fail-open. All interlocks latched. Manual reset required. Entry: any ESD interlock condition.

4. SETPOINTS TABLE

Tag	Parameter	Value	Units	Page Ref
PT-101	High High Trip (HH)	150	PSI	This page
PT-101	High Alarm (H)	130	PSI	This page
PT-101	Low Alarm (L)	20	PSI	This page
FT-201	Low Flow Trip (LL)	10	L/min	This page
FT-201	Low Alarm (L)	20	L/min	This page
FT-201	Flow Control SP	80	L/min	This page
LT-301	Low Low Trip (LL)	15	%	This page
LT-301	Low Alarm (L)	25	%	This page
TT-401	High High Trip (HH)	85	°C	This page
TT-401	High Alarm (H)	75	°C	This page
FV-301	Fail Position	Open	—	This page

5. INTERLOCKS

The following safety interlocks shall be implemented in the PLC. All safety interlocks require manual reset by the operator after the initiating condition has cleared and the root cause has been investigated.

Interlock IL-101 — High High Discharge Pressure Trip

Field	Content
Cause (Initiating Condition)	PT-101 discharge pressure exceeds 150 PSI for more than 3 seconds
Action	Stop pump P-101 AND stop pump P-102 AND close XV-201 AND generate alarm 'HIGH HIGH PRESSURE TRIP' on HMI
Reset Condition	PT-101 pressure falls below 120 PSI AND operator has investigated root cause
Reset Method	Manual reset — operator must press RESET on HMI Panel
Priority	Safety Critical

Interlock IL-201 — Low Low Flow Trip

Field	Content
Cause (Initiating Condition)	FT-201 flow rate falls below 10 L/min for more than 5 seconds while pump running
Action	Stop pump P-101 AND auto-start standby pump P-102 AND generate alarm 'LOW FLOW TRIP'
Reset Condition	FT-201 flow rate above 20 L/min AND P-102 confirmed running

Field	Content
Reset Method	Automatic reset after 30 seconds if P-102 restores flow
Priority	High

Interlock IL-301 — Low Low Suction Level Trip

Field	Content
Cause (Initiating Condition)	LT-301 suction vessel level falls below 15% for more than 10 seconds
Action	Stop pump P-101 AND stop pump P-102 AND close XV-101 AND generate alarm 'LOW SUCTION LEVEL'
Reset Condition	LT-301 level above 25% AND operator confirms feed restored
Reset Method	Manual reset — operator must press RESET on HMI Panel
Priority	Safety Critical

6. ALARMS

Tag	Type	Setpoint	Units	Priority	Operator Response
PT-101-HH	HH Trip	150	PSI	Priority 1	Stop all pumps. Close XV-201. Investigate overpressure cause. Do not restart until root cause confirmed.
PT-101-H	H Alarm	130	PSI	Priority 2	Reduce flow setpoint on FIC-201 by 20%. Monitor PT-101 trend. Prepare for potential trip.
PT-101-L	L Alarm	20	PSI	Priority 3	Check pump operation. Verify XV-201 and XV-201 are fully open. Inspect for downstream blockage.
FT-201-L	L Alarm	20	L/min	Priority 2	Check FV-301 position. Verify pump P-101 is running at correct speed. Check for line blockage.
LT-301-L	L Alarm	25	%	Priority 2	Contact feed operations to restore suction vessel level. Prepare for pump shutdown if level continues falling.
TT-401-H	H Alarm	75	°C	Priority 2	Reduce pump load or switch to P-102. Check bearing lubrication. Schedule P-101 maintenance inspection.

7. STARTUP SEQUENCE

The following sequence shall be executed by the PLC when the operator initiates STARTUP from the HMI. Each step includes a verification check and a maximum timeout. If a step times out without confirmation, the sequence shall abort and generate an alarm.

Step	Action	Details / Verification
Step 1	Check Permissives	Verify: LT-301 level > 25%, PT-101 < 100 PSI, no active ESD conditions. Timeout: N/A (permissive check only). Abort if any condition fails.
Step 2	Open Suction Valve XV-101	Command XV-101 to OPEN. Verify: XV-101 open limit switch confirmed within 15 seconds. Timeout: 15 seconds. Abort sequence and alarm if not confirmed.
Step 3	Prime Pump P-101	Wait 5 seconds after XV-101 confirmed open to allow pump casing to fill.
Step 4	Start Duty Pump P-101	Command P-101 motor starter to START. Verify: P-101 running feedback confirmed within 10 seconds. Timeout: 10 seconds.
Step 5	Open Discharge Valve XV-201	Command XV-201 to OPEN. Verify: XV-201 open limit switch confirmed within 15 seconds. Timeout: 15 seconds.
Step 6	Confirm Flow Established	Verify: FT-201 > 20 L/min within 30 seconds of XV-201 open. Timeout: 30 seconds. If no flow — stop P-101, close XV-201, alarm 'NO FLOW ON STARTUP'.
Step 7	Enable Flow Controller	Enable FIC-201 in AUTO mode with setpoint 80 L/min. Ramp FV-301 from current position to target over 10 seconds to avoid water hammer.
Step 8	Enable Standby Auto-Start	Enable P-102 auto-start logic. P-102 will auto-start if P-101 trips and flow drops below 15 L/min.
Step 9	Startup Complete	Set operating mode to RUNNING. Log startup timestamp to audit trail. Generate event: 'STARTUP COMPLETE — PS-01 IN NORMAL OPERATION'.

8. NORMAL SHUTDOWN SEQUENCE

Step	Action	Details / Verification
Step 1	Disable Auto-Start	Disable P-102 auto-start logic.
Step 2	Ramp Down Flow	Reduce FIC-201 setpoint from 80 to 20 L/min over 15 seconds.
Step 3	Close FV-301	Switch FIC-201 to MANUAL. Close FV-301 fully over 10 seconds.
Step 4	Close XV-201	Command XV-201 to CLOSE. Verify closed within 15 seconds.
Step 5	Stop Pump P-101	Command P-101 to STOP after XV-201 confirmed CLOSED. Verify P-101 stop feedback within 5 seconds.
Step 6	Close XV-101	Command XV-101 to CLOSE. Verify closed within 15 seconds.
Step 7	Shutdown Complete	Set mode to STOPPED. Log shutdown timestamp.

9. EMERGENCY SHUTDOWN (ESD)

The Emergency Shutdown sequence is triggered automatically by any ESD interlock condition or manually by the operator pressing the ESD pushbutton on the panel.

Step	Action	Details / Verification
ESD-1	Stop all pumps	De-energise P-101 AND P-102 motor starters simultaneously.
ESD-2	Close XV-201	Command XV-201 to CLOSE immediately. Hardwired signal bypasses PLC.
ESD-3	Close XV-101	Command XV-101 to CLOSE to isolate suction.
ESD-4	De-energise FV-301	FV-301 de-energised — moves to fail-open position automatically.
ESD-5	Latch ESD state	Set ESD latch bit. All normal control commands ignored until reset.
ESD-6	Generate alarms	Activate all ESD alarms on HMI and initiate audible alarm.

NOTE: Manual reset of ESD latch requires operator to: (1) confirm all ESD conditions cleared, (2) investigate and document root cause, (3) press RESET on the PLC panel.